

Problem 9.10

Calculate the nominal rate of discount convertible monthly that is equivalent to a nominal rate of interest of 18.9% per year convertible monthly.

Solution

Let

$$i^{(12)} = 0.189$$

be the nominal annual rate of interest convertible monthly. The corresponding monthly effective interest rate is

$$\frac{i^{(12)}}{12} = \frac{0.189}{12}.$$

For equivalent nominal rates convertible monthly, the monthly interest rate and the monthly discount rate must satisfy

$$1 + \frac{i^{(12)}}{12} = \frac{1}{1 - \frac{d^{(12)}}{12}}.$$

Equivalently,

$$\frac{d^{(12)}}{12} = \frac{\frac{i^{(12)}}{12}}{1 + \frac{i^{(12)}}{12}}.$$

Therefore,

$$d^{(12)} = 12 \left(\frac{\frac{0.189}{12}}{1 + \frac{0.189}{12}} \right).$$

Now compute:

$$d^{(12)} = 12 \left(\frac{0.01575}{1.01575} \right) \approx 0.18606940.$$

Thus, the equivalent nominal annual rate of discount convertible monthly is

$$d^{(12)} \approx 18.606940\%.$$